TECHNICAL REPORT

IoT-Based Network Rack Tamper Detection and Vibration Alert System

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1. Project Title

IoT-Based Network Rack Tamper Detection and Vibration Alert System

2. Abstract

The IoT-Based Network Rack Tamper Detection and Vibration Alert System is a low-cost security system designed to detect unauthorized physical movement, vibration, or tampering of a network rack. The system uses an SW-420 vibration sensor to detect vibration. An ESP8266 NodeMCU acts as the main controller and connects the system to the Internet through Wi-Fi. When vibration is detected, the NodeMCU identifies the condition as TAMPER DETECTED.

The system provides multiple alerts: a physical LED indication, a buzzer alarm, a Blynk mobile notification, and a Blynk dashboard status. When no vibration is detected, the system displays SAFE, turns the LED and buzzer OFF, and continuously monitors the rack.

3. Problem Statement

Network racks contain important networking equipment such as routers, switches, servers, patch panels, network cables, firewalls, and other communication equipment. Unauthorized physical access or accidental

movement can cause equipment damage, cable disconnection, network interruption, and physical security issues. This project provides a simple IoT-based solution that detects vibration, provides a local alarm, sends a remote alert, and displays rack status on a mobile device.

4. Objectives

- Detect physical vibration or tampering of a network rack.
- Use an ESP8266 NodeMCU to process the sensor signal.
- Display the rack condition as SAFE or TAMPER DETECTED.
- Turn ON a physical LED when tampering is detected.
- Activate a buzzer during tampering.
- Send a notification to a mobile phone using Blynk.
- Monitor the rack remotely through Wi-Fi.

5. System Architecture

Network Rack
↓
SW-420 Vibration Sensor
↓
ESP8266 NodeMCU
├── D6 → Red LED
├── D7 → Buzzer
└── Wi-Fi → Blynk Cloud → Mobile App

6. Working Principle

6.1 Normal Condition

When the network rack is stationary, the SW-420 does not detect vibration. The NodeMCU reads the sensor as LOW, the rack status is SAFE, and the LED and buzzer remain OFF.

6.2 Tampering Condition

When the rack is moved or vibrated, the SW-420 detects vibration and sends a signal to the NodeMCU. The NodeMCU identifies the condition as TAMPER DETECTED, turns ON the LED and buzzer, updates Blynk, and triggers a mobile notification.

7. Hardware Components Required

No.	Component	Quantity	Purpose
1	ESP8266 NodeMCU	1	Main IoT controller
2	SW-420 vibration sensor	1	Detect vibration
3	Red LED	1	Local tamper indication
4	220 Ω resistor	1	LED current limiting
5	2-pin buzzer	1	Audible alert
6	Breadboard	1	Circuit assembly
7	Jumper wires	Several	Connections
8	USB data cable	1	Programming and power
9	Laptop/PC	1	Programming
10	Android phone	1	Blynk monitoring

8. Software Required

- Arduino IDE for programming the NodeMCU.
- ESP8266 board package – required to program the ESP8266.
- Blynk Library – for communication between NodeMCU and Blynk.
- Blynk IoT mobile application – for monitoring and notifications.
- Wi-Fi Internet connection – for Blynk cloud communication.

9. Arduino IDE Installation

1. Download and install Arduino IDE on the laptop.

2. Open Arduino IDE.
3. If Arduino IDE is already installed, no reinstallation is necessary.

10. USB Driver and COM Port

4. Connect the NodeMCU to the laptop using a USB data cable.
5. Open Device Manager.
6. Open Ports (COM & LPT).
7. Confirm that the NodeMCU appears as a COM port.
8. In this project, the NodeMCU was available on COM3.
9. If the COM port is already visible, the USB driver is already installed, and no additional driver installation is required.

11. ESP8266 Board Installation

10. Open Arduino IDE.
11. Go to File → Preferences.
12. Add the ESP8266 board package URL in Additional Boards Manager URLs.
13. Click OK.
14. Go to Tools → Board → Boards Manager.
15. Search for ESP8266.
16. Install ESP8266 by ESP8266 Community.

12. Select NodeMCU Board and Port

17. Go to Tools → Board → ESP8266 Boards.
18. Select NodeMCU 1.0 (ESP-12E Module).
19. Go to Tools → Port.
20. Select COM3.

13. Blynk Library Installation

21. In Arduino IDE go to Sketch → Include Library → Manage Libraries.
22. Search for Blynk.
23. Install the Blynk library by Volodymyr Shymanskyy.

14. Blynk IoT Configuration

Template ID: TMPL39DzPVbuT

Template Name: Network Rack Tamper Detection And Vibration Alert

15. Blynk DataStream

Virtual Pin	Name	Purpose
V0	Rack Status	SAFE / TAMPER DETECTED
V1	Rack LED	LED status
V2	Vibration	Sensor value

16. Blynk Event Configuration

- Event Name: Rack Tamper
- Event Code: rack_tamper
- Notification Message: Network Rack Tampering Detected!
- Enable push notifications for the event.

17. Mobile Notification Permission

24. Open Android Settings.
25. Go to Apps → Blynk IoT → Notifications.
26. Enable Allow notifications.
27. Make sure notification categories are enabled if displayed.
28. Disable any phone setting that blocks Blynk notifications, such as an applicable Do Not Disturb rule.

18. Hardware Wiring

18.1 SW-420 Vibration Sensor

SW-420 VCC → NodeMCU 3V/3V3
SW-420 GND → NodeMCU GND
SW-420 DO → NodeMCU D5

18.2 Red LED

NodeMCU D6 → 220 Ω resistor → LED long leg (+)
LED short leg (−) → NodeMCU GND

The 220 Ω resistor is required to limit LED current. Do not connect the LED directly to the GPIO pin.

18.3 2-Pin Buzzer

Buzzer + → NodeMCU D7
Buzzer − → NodeMCU GND

The positive pin is the pin marked '+'. The other pin is negative.

19. Final Pin Configuration

NodeMCU Pin	Connected Component
3V/3V3	SW-420 VCC
GND	SW-420 GND
D5	SW-420 DO
D6	220 Ω resistor → LED
D7	Buzzer +
GND	LED − and Buzzer −

20. Wire Color Recommendation

Wire colors do not affect electrical operation. For a clean project, red can be used for 3V/3V3, black for GND, yellow for sensor DO, green/blue for the LED signal, and blue for the buzzer signal.

21. Sensor Testing Before Blynk

The SW-420 can first be tested independently using the following code:

```
void setup() {
  Serial.begin(115200);
  pinMode(D5, INPUT);
}
```

```
void loop() {
  int vibration = digitalRead(D5);

  if (vibration == HIGH) {
    Serial.println("TAMPER DETECTED");
  } else {
    Serial.println("SAFE");
  }

  delay(500);
}
```

22. Sensor Test Procedure

29. Connect the NodeMCU to the laptop.
30. Select NodeMCU 1.0 (ESP-12E Module).
31. Select COM3.
32. Paste the sensor test code.
33. Click Verify.
34. Click Upload after compilation succeeds.
35. Open Tools → Serial Monitor.
36. Set the baud rate to 115200.
37. With the sensor stationary, SAFE should appear.
38. When the sensor is gently shaken, TAMPER DETECTED should appear.

23. Complete Final Arduino Code

```
#define BLYNK_TEMPLATE_ID "TMPL39DzPVbuT"
#define BLYNK_TEMPLATE_NAME "Network Rack Tamper Detection
And Vibration Alert"
#define BLYNK_AUTH_TOKEN "YOUR_AUTH_TOKEN"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
```

```
// Wi-Fi details
char ssid[] = "YOUR_WIFI_NAME";
char pass[] = "YOUR_WIFI_PASSWORD";

// Hardware pins
#define VIBRATION_PIN D5
#define LED_PIN D6
#define BUZZER_PIN D7

BlynkTimer timer;

// Stores the current tamper state
bool tamperState = false;

void checkVibration()
{
 int vibration = digitalRead(VIBRATION_PIN);

 // TAMPER DETECTED
 if (vibration == HIGH)
 {
  digitalWrite(LED_PIN, HIGH);
  digitalWrite(BUZZER_PIN, HIGH);

  Blynk.virtualWrite(V0, "TAMPER DETECTED");
  Blynk.virtualWrite(V1, 1);
  Blynk.virtualWrite(V2, 1);

  // Send one notification for this incident
  if (tamperState == false)
  {
   Blynk.logEvent(
    "rack_tamper",
    "Network Rack Tampering Detected!"
   );

   tamperState = true;
```

```
    }
  }

  // SAFE
  else
  {
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);

    Blynk.virtualWrite(V0, "SAFE");
    Blynk.virtualWrite(V1, 0);
    Blynk.virtualWrite(V2, 0);

    // Allow another alert for a future incident
    tamperState = false;
  }
}

void setup()
{
  Serial.begin(115200);

  pinMode(VIBRATION_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LED_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  timer.setInterval(200L, checkVibration);
}

void loop()
{
  Blynk.run();
```

```
  timer.run();
}
```

24. Important Code Configuration

- Replace YOUR_AUTH_TOKEN with the Blynk Auth Token for your device.
- Replace YOUR_WIFI_NAME with the Wi-Fi network name.
- Replace YOUR_WIFI_PASSWORD with the Wi-Fi password.
- Do not share the Auth Token or Wi-Fi password.
- Do not change the Template ID, Template Name, D5, D6, or D7 values unless the physical wiring is changed.

25. Code Function Summary

Code element	Function
BLYNK_TEMPLATE_ID	Identifies the Blynk template
BLYNK_AUTH_TOKEN	Authenticates the device
D5	Receives SW-420 vibration signal
D6	Controls red LED
D7	Controls buzzer
V0	Displays rack status
V1	Reports LED status
V2	Reports vibration status
Blynk.logEvent()	Triggers the Blynk notification event
tamperState	Prevents repeated notifications during one continuous incident

26. Final Program Upload Procedure

39. Save the final code.
40. Select Tools → Board → NodeMCU 1.0 (ESP-12E Module).
41. Select Tools → Port → COM3.
42. Click Verify and wait for successful compilation.
43. Click Upload.
44. Wait for Done uploading.

45. Wait for the NodeMCU to reconnect to Wi-Fi and Blynk.

27. Testing Procedure

27.1 Normal Condition

Keep the sensor stationary. Expected: Blynk = SAFE, physical LED = OFF, buzzer = OFF.

27.2 Tamper Condition

Gently shake or tap the SW-420. Expected: Blynk = TAMPER DETECTED, physical LED = ON, buzzer = ON, and a Blynk mobile notification.

27.3 Recovery

Stop the vibration. Expected: Blynk returns to SAFE, LED turns OFF, and buzzer turns OFF.

27.4 New Incident

After the system returns to SAFE, trigger vibration again. The system is designed to generate a new notification for the new incident.

28. Notification Flow

SW-420 → NodeMCU → Wi-Fi → Blynk Cloud → Blynk Event → Push Notification → Android Phone

A SIM card or GSM module is not required for Blynk push notifications. Internet connectivity is required.

29. Demonstration Procedure

46. Show the SW-420 vibration sensor.
47. Show the ESP8266 NodeMCU.
48. Show the red LED and 220 Ω resistor.
49. Show the 2-pin buzzer.
50. Show the Blynk dashboard.

51. Keep the sensor still and show SAFE.
52. Gently shake the sensor.
53. Show TAMPER DETECTED on Blynk.
54. Show the LED turning ON.
55. Show the buzzer sounding.
56. Show the mobile notification.
57. Stop vibration and show the return to SAFE.

30. Removing the Hardware After Demonstration

58. Stop the demonstration.
59. Disconnect the NodeMCU USB cable.
60. Remove the SW-420 wires.
61. Remove the LED wires.
62. Remove the buzzer wires.
63. Store the components safely.

The Arduino IDE, ESP8266 board package, Blynk library, and Blynk configuration do not need to be reinstalled for every demonstration.

31. Reassembly for Another Demonstration

Reconnect the same pins: SW-420 VCC → 3V3, SW-420 GND → GND, SW-420 DO → D5, D6 → 220 Ω → LED, LED negative → GND, D7 → buzzer positive, buzzer negative → GND. The previously uploaded program remains stored on the NodeMCU, so it normally does not need to be uploaded again.

32. Advantages

- Low-cost implementation.
- Simple hardware.
- Easy to demonstrate.
- Wireless monitoring.
- Mobile notification.
- Local LED indication.

- Audible alarm.
- Real-time monitoring.
- Easy to expand.

33. Limitations

- The SW-420 detects vibration rather than identifying the person causing it.
- Accidental vibration can trigger an alert.
- Strong nearby vibration can trigger the sensor.
- Sensor sensitivity may need adjustment.
- Blynk cloud notification requires Internet connectivity.
- The SW-420 cannot distinguish between different types of physical movement.

34. Future Enhancements

- Add a magnetic reed switch for rack-door opening detection.
- Add camera integration for image capture.
- Add GSM/SMS support for Internet-independent alerts.
- Use multiple sensors at different rack locations.
- Store tamper events with timestamps in a cloud database.
- Add battery backup.
- Add email alerts.
- Add tamper history and analytics.

35. Testing Table

Test	Expected Result
Sensor stationary	SAFE
Sensor vibrated	TAMPER DETECTED
Tamper detected	LED ON
Tamper detected	Buzzer ON
Tamper detected	Blynk status changes
Tamper detected	Mobile notification
Vibration stops	SAFE

Vibration stops	LED OFF
Vibration stops	Buzzer OFF
New vibration	New tamper notification

36. Troubleshooting

NodeMCU not detected

- Check USB cable.
- Check USB driver.
- Check COM port.
- Try another USB port.

No COM port

- Try another USB data cable.
- Check Device Manager.
- Check the ESP8266 USB driver.

Serial Monitor shows ?? symbols

Set Serial Monitor to 115200 baud because the program uses Serial.begin(115200).

LED does not turn ON

- Check D6.
- Check 220 Ω resistor.
- Check LED polarity.
- Check GND.

Buzzer does not sound

- Check buzzer + to D7.
- Check buzzer − to GND.
- If it remains silent, the buzzer may be passive and may require a tone signal rather than HIGH/LOW control.

Blynk does not show status

- Check Wi-Fi.
- Check Auth Token.
- Check Template ID and Template Name.
- Check V0, V1, and V2 datastreams.
- Check the NodeMCU Internet connection.

Mobile notification does not arrive

- Check Event Code = rack_tamper.
- Enable push notifications in the Blynk event.
- Enable Android notification permission.
- Check Internet connectivity.
- Confirm the correct Blynk account is being used.

37. Security Considerations

- Keep the Blynk Auth Token private.
- Keep the Wi-Fi password private.
- Do not publish credentials in screenshots, reports, or source repositories.

38. Final Project Flow

START
↓
Power NodeMCU
↓
Connect to Wi-Fi
↓
Connect to Blynk
↓
Read SW-420
├── No vibration → SAFE → LED OFF → Buzzer OFF
│

└── Vibration → TAMPER DETECTED → LED ON → Buzzer ON

↓

Blynk Event

↓

Mobile Alert

↓

Repeat

39. Final Output

The completed system provides three levels of tamper indication: a local visual alert through the red LED, a local audible alert through the buzzer, and a remote alert through Blynk mobile notification. The system provides a simple, low-cost IoT approach for monitoring physical vibration or tampering of a network rack.

40. Overall Explanation

Our project is an IoT-based Network Rack Tamper Detection and Vibration Alert System. We use an SW-420 vibration sensor to detect physical movement or vibration of the rack. The sensor is connected to an ESP8266 NodeMCU through D5. When vibration is detected, the NodeMCU turns ON a red LED connected through a 220-ohm resistor and activates a buzzer through D7. At the same time, the NodeMCU sends the status through Wi-Fi to the Blynk Cloud. Blynk updates the dashboard to TAMPER DETECTED and sends a push notification to the mobile application. When the vibration stops, the system returns to SAFE and turns the LED and buzzer OFF.

41. Key Technical Configuration

Parameter	Configuration
Controller	ESP8266 NodeMCU
Sensor	SW-420 vibration sensor
Sensor Pin	D5

LED Pin	D6
Buzzer Pin	D7
LED Resistor	220 Ω
Cloud Platform	Blynk IoT
Rack Status	V0
Rack LED	V1
Vibration	V2
Event Code	rack_tamper
Communication	Wi-Fi
Mobile Alert	Blynk Push Notification